Anomaly Segmentation: Comprehensive Road Scene Understanding for Autonomous Driving

Group 08

Simone Duma s353855
Giovanni Indiano s357942
Paolo Malugani s359857
Fabio Stradiotti s359415

Abstract

Perception systems in autonomous driving are based on deep neural networks for accurate scene understanding. Despite this, they often fail to identify out-of-distribution (OoD) objects not seen before, which represents a critical safety risk in real environments. In this work, we use the foundation-model-based Encoder-only Mask Transformer (EoMT) for both image segmentation and anomaly detection. We first map and fine-tune a COCO-pretrained EoMT on the Cityscapes dataset to bridge the domain gap between generic visual priors and specialized urban environments. Next, we consider various post-hoc OoD detection methods on the SegmentMeIfYouCan (SMIYC) and Fishyscapes benchmarks. The experiments show that, although region-level metrics like Rejected by All (RbA) theoretically align with mask classification, they are characterized by overconfidence and high false positive rates. Instead, traditional confidence scores such as Maximum Softmax Probability (MSP) and MaxEntropy achieve better results. In particular, MaxEntropy achieves 91.74% AuPRC and 2.60% FPR95 on SMIYC RA-21. In the end, in order to improve the model's calibration, we apply temperature scaling. The source code for this project is available at: `https://github.com/strafabio1/AnomalySegmentation_Project`

1. Introduction

Driven by large-scale datasets like Cityscapes [5] and COCO [9], semantic and panoptic segmentation-related studies evolved to the mask classification paradigm (such as MaskFormer and Mask2Former [3, 4]) which outperform the previous pixel-level predictions like ERFNet [12]. Moreover, foundation models like DINOv2 [11] provide robust, generalizable features. These backbones are used by architectures such as the Encoder-only Mask Transformer (EoMT) [8] for highly efficient mask classification. Despite this, detecting Out-of-Distribution (OoD) objects remains a critical challenge for autonomous driving safety. Anomaly segmentation addresses this by isolating unknown regions. While early methods relied on Maximum Softmax Probability (MSP) or MaxLogit, modern approaches use mask classification; for instance, Rejected by All (RbA) [10] identifies anomalies as regions explicitly rejected by all known classifiers. In this project, we investigate EoMT's OoD generalization for road scenes. To bridge the domain gap, we map and fine-tune a COCO-pretrained EoMT on Cityscapes, subsequently evaluating advanced post-hoc methods. To recap, our work consists of three steps:

1. We use both strategic class mapping and fine-tuning to adapt a foundation-model architecture to urban scenes.
2. We critically evaluate post-hoc scoring functions, demonstrating that standard baselines (e.g., MSP, MaxEntropy) offer better calibration compared to region-based approaches like RbA.
3. We apply temperature scaling to optimize predictive confidence and finally validate our pipeline on Fishyscapes [1] and SMIYC [2].

2. Methodology

2.1. Semantic and Panoptic Segmentation

In order to highlight the differences between the two pre-trained EoMT models, we compare their predictions on images from the Cityscapes validation set. The primary distinction concerns the specific tasks the models were originally trained for. The EoMT model pre-trained on the COCO dataset performs panoptic segmentation, which makes it possible to distinguish between individual instances of the same category, assigning them unique mask IDs (e.g., separating adjacent vehicles or pedestrians). On the contrary, the Cityscapes-pre-trained model produces semantic segmentation masks, where adjacent objects belonging to the same class are merged into a single contiguous region. The misalignment of label spaces represents the second major difference. The Cityscapes-pre-trained model shows high sensitivity to urban-specific infrastructure (e.g., traffic signs, poles, and sidewalks). On the other hand, since these categories are not represented with the same label granularity in the COCO taxonomy, the COCO-trained model often fails to recognize them as distinct urban entities, grouping them into broader background classes like *building* or *vegetation*. Indeed, a coherent class mapping strategy is needed.

2.2. Mapping the COCO Classes

The first problem we have to address to quantitatively evaluate the COCO-pretrained EoMT model on the Cityscapes validation set is to solve the structural mismatch between their taxonomies (133 generic classes in COCO, 19 semantic categories in Cityscapes). We face this problem with the following mapping function

$$f_{map}(c) = \begin{cases} c_{city} & \text{if } c \text{ is an exact semantic match} \\ & \text{(e.g., } \textit{car} \rightarrow \textit{car}\text{)} \\ c_{city} & \text{if } c \in \mathcal{S}_{city} \\ & \text{(aggregation of } \textit{stuff} \text{ classes)} \\ c_{city} & \text{if } c \text{ is a logical urban equivalent} \\ & \text{(e.g., } \textit{stop sign} \rightarrow \textit{traffic sign}\text{)} \\ \emptyset & \text{otherwise (e.g. animals)} \end{cases} \tag{1}$$

For example, the subset aggregation $\mathcal{S}_{\text{vegetation}}$ merges COCO classes like *tree*, *grass*, and *flower*. Structural materials (*wall-brick*, *wall-stone*) are mapped to $\mathcal{S}_{\text{wall}}$. Since the mapped COCO baseline cannot provide reliable predictions for every Cityscapes class, we restrict our common evaluation space to a 17-class subset, excluding the unsupported classes *rider* and *pole*, in order to avoid unfair penalization during metric computation.

2.3. Fine-tuning Strategy

To reduce the domain gap while preserving the prior knowledge of COCO-pre-trained model, we fine-tune the model on the Cityscapes training set using a progressive unfreezing strategy. We initialize the network with the frozen DINOv2 encoder and update only the final prediction head, which allows the segmentation head to adapt to the 19 Cityscapes classes. Subsequently, we gradually unfreeze the deepest encoder blocks to capture detailed urban patterns. Specifically, we evaluate and compare specific configurations where the last one, two, and three encoder blocks are unfrozen. This approach allows us to quantify the gain on our restricted 17-class subset and directly assess the overall 19-class domain competence against the native Cityscapes pre-trained model. In the end, we choose the configuration with the last three unfrozen blocks, trained for 20 epochs using Automatic Mixed Precision (AMP) to optimize computational efficiency. Moreover, to be aligned with the original EoMT training dynamics (which require an effective batch size of 16 [8]) under tight memory constraints, we employ a localized batch size of 8 combined with a gradient accumulation factor of 2.

2.4. Anomaly Segmentation and Post-Hoc Methods

We formulate anomaly segmentation as an Out-of-Distribution (OoD) detection task, by using post-hoc uncertainty estimation methods rather than modifying the underlying training paradigm. For our pixel-level baseline, ERFNet, the network predicts a dense logit vector $\mathbf{f}(\mathbf{x}) \in \mathbb{R}^{|\mathcal{K}|}$ at each spatial location $\mathbf{x}$, where $\mathcal{K}$ denotes the set of known in-distribution classes. Let $p(y = c|\mathbf{x})$ denote the standard softmax probability for class $c \in \mathcal{K}$. We derive pixel-independent anomaly scores using three established criteria:

- Maximum Softmax Probability (MSP) [6]: Scores the anomaly based on the complement of the highest confidence. To calibrate the predictive distribution specifically for this metric, we introduce Temperature Scaling (T) to the logits:

$$\text{MSP}(\mathbf{x}) = 1 - \max_{c \in \mathcal{K}} \text{softmax}_c(\mathbf{f}(\mathbf{x})/T) \tag{2}$$

- MaxLogit [7]: Leverages the negative of the maximum raw logit:

$$\text{MaxLogit}(\mathbf{x}) = -\max_{c \in \mathcal{K}} f_c(\mathbf{x}) \tag{3}$$

- MaxEntropy: Evaluates the Shannon entropy of the standard predictive distribution:

$$\text{MaxEntropy}(\mathbf{x}) = -\frac{1}{\ln(|\mathcal{K}|)} \sum_{c \in \mathcal{K}} p(y = c|\mathbf{x}) \ln p(y = c|\mathbf{x}) \tag{4}$$

In order to extend these post-hoc metrics to mask-classification architectures, such as EoMT, we need to aggregate the N object queries into a dense spatial representation. Following [10], let F be the pixel features and Q the object queries. The mask predictions $M \in [0,1]^{N \times H \times W}$ are generated via $M = \sigma(QF^T)$, where σ denotes the Sigmoid activation function. Concurrently, the classifier outputs the query-level class logits $L \in \mathbb{R}^{N \times |\mathcal{K}|}$. We compute two types of dense spatial maps for each class $c \in \mathcal{K}$: the aggregated final probability score $F_c(\mathbf{x})$ and the aggregated logit score $Z_c(\mathbf{x})$. To properly apply Temperature Scaling (T) in this paradigm, the scaled query probabilities $P^{(T)}_{n,c} = \text{softmax}_c(L_n/T)$ are computed before aggregation. The dense maps are defined as:

$$F_c(\mathbf{x}) = \sum_{n=1}^{N} P^{(T)}_{n,c} \cdot M_n(\mathbf{x}), \quad Z_c(\mathbf{x}) = \sum_{n=1}^{N} L_{n,c} \cdot M_n(\mathbf{x}) \tag{5}$$

With these aggregated maps, the standard post-hoc methods are seamlessly adapted:

- MSP: Applies the complement of the highest aggregated probability:

$$\text{MSP}(\mathbf{x}) = 1 - \max_{c \in \mathcal{K}} F_c(\mathbf{x}) \tag{6}$$

- MaxLogit: Evaluates the negative of the maximum aggregated logit:

$$\mathrm{MaxLogit}(\mathbf{x}) = -\max_{c \in \mathcal{K}} Z_c(\mathbf{x}) \tag{7}$$

- MaxEntropy: The aggregated probabilities $F_c(\mathbf{x})$ are first normalized into a valid distribution $\hat{p}_c(\mathbf{x}) = F_c(\mathbf{x}) / \sum_{j \in \mathcal{K}} F_j(\mathbf{x})$, over which the standard Shannon entropy is computed.

Additionally, we evaluate the Rejected by All (RbA) metric [10], which is specifically tailored for mask-based paradigms. It explicitly measures the extent to which all known-class classifiers reject a given region by applying a non-linear hyperbolic tangent transformation to the unscaled aggregated probabilities $F_c(\mathbf{x})$:

$$\mathrm{RbA}(\mathbf{x}) = -\sum_{c \in \mathcal{K}} \tanh(F_c(\mathbf{x})) \tag{8}$$

2.5. Evaluation Metrics

For the Semantic Segmentation task, performance on in-distribution Cityscapes data is evaluated using the mean Intersection over Union (mIoU). It measures the average overlap between the predicted segmentation mask and the ground truth across all C valid classes:

$$\mathrm{mIoU} = \frac{1}{C} \sum_{c=1}^{C} \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{9}$$

where TP_c, FP_c, and FN_c are the true positives, false positives, and false negatives for class c, respectively. For the Anomaly Segmentation task, pixels are binary-classified as *anomalous* (positive class = 1) or *normal* (negative class = 0), then anomaly maps are evaluated using two threshold-independent metrics:

- Area Under the Precision-Recall Curve (AuPRC): Summarizes the trade-off between precision ($\frac{\mathrm{TP}}{\mathrm{TP+FP}}$) and recall ($\frac{\mathrm{TP}}{\mathrm{TP+FN}}$) across all possible anomaly score thresholds. It is highly robust to the severe class imbalance typical of anomaly datasets, where normal pixels vastly outnumber anomalous ones.
- False Positive Rate at 95% TPR (FPR95): Evaluates the probability that a normal pixel is mistakenly classified as an anomaly when the decision threshold is set to correctly identify 95% of the true anomalies (True Positive Rate = 0.95).

3. Experimental Results

3.1. Implementation Details

We evaluate all models on the Cityscapes validation set considering the mIoU. To reduce the impact of label space misalignment, zero-shot and mapped comparisons use a restricted 17-class subset, while progressive unfreezing works

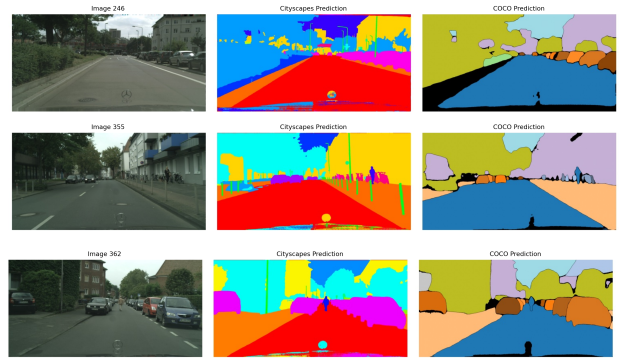

Figure 1. Qualitative zero-shot comparison. Left to right: input, semantic predictions (native Cityscapes), panoptic predictions (native COCO). Note the semantic grouping vs. instance separation and the vocabulary mismatch.

on the full 19-class taxonomy. Out-of-distribution (OoD) anomaly segmentation is benchmarked on five standard datasets (SMIYC RA-21, SMIYC RO-21, FS L&F, FS Static, Road Anomaly) using AuPRC and FPR95.

3.2. Semantic and Panoptic Evaluation

Table 1 reports evaluations on the 17-class subset (excluding *rider* and *pole* due to the COCO taxonomy mismatch). The native Cityscapes model establishes an 83.60% mIoU upper bound. The zero-shot *Mapped COCO* baseline yields 62.92% mIoU: while it successfully parses broad classes (*road*, *sky*), it collapses on urban-specific infrastructure like *traffic sign* (4.43%), which it visually absorbs into broader background regions (see COCO prediction in Figure 1). Targeted fine-tuning (*Fine-tuned COCO*) effectively mitigates this domain gap, raising the global score to 79.20% and completely recovering performance on critical categories (e.g., *traffic sign* reaching 77.21%). Table 2 shows the progressive unfreezing ablation on the full 19-class set. Sequentially unfreezing deeper encoder layers provides consistent gains, peaking at 76.13% mIoU with three unfrozen blocks. The *rider* category, masked during zero-shot evaluation, progresses from 31.96% (head-only update) to a robust 61.27% IoU in the final configuration.

3.3. Out-of-Distribution Anomaly Segmentation

Table 3 highlights a strong performance disparity between architectural paradigms. The pixel-based ERFNet performs poorly (max 38.32% AuPRC on SMIYC RA-21), whereas the domain-aligned mask architecture (*EoMT Cityscapes*) successfully isolates anomalous objects, achieving 89.76% AuPRC and an exceptional 0.44% FPR95 on SMIYC RO-21 with MSP.

However, the efficacy of post-hoc adjustments heavily depends on the architecture. While *MaxLogit* improves ERFNet, it causes severe metric deterioration on EoMT (e.g., SMIYC RO-21 drops to 47.56% AuPRC), proving that raw mask logits lack energy calibration. Similarly, *RbA*

Table 1. Per-class IoU (%) on the Cityscapes validation set (17-class subset).

Model	Roa	Sid	Bui	Wal	Fen	TLi	TSi	Veg	Ter	Sky	Per	Car	Tru	Bus	Tra	Mot	Bic	mIoU
Cityscapes	98.42	87.62	94.66	66.23	66.02	77.53	83.01	93.38	67.08	95.68	86.58	95.69	81.85	90.49	77.74	76.28	82.94	83.60
Mapped COCO	94.10	66.58	89.36	46.91	49.80	62.80	4.43	89.44	58.28	90.73	76.35	84.51	29.50	71.77	13.70	66.80	74.64	62.92
Fine-tuned COCO	97.97	83.71	93.10	61.99	62.75	69.87	77.21	92.26	65.61	94.64	83.17	94.56	76.43	85.15	68.15	61.39	78.43	79.20

Table 2. Ablation of progressive unfreezing on the full 19-class Cityscapes validation set.

Model	Roa	Sid	Bui	Wal	Fen	Pol	TLi	TSi	Veg	Ter	Sky	Per	Rid	Car	Tru	Bus	Tra	Mot	Bic	mIoU
Cityscapes	98.40	87.36	94.15	66.07	65.49	71.04	75.00	82.13	93.02	66.60	95.54	85.38	71.11	95.54	81.78	90.32	77.36	74.35	81.27	81.68
Head only	97.70	81.89	91.84	59.65	55.31	52.94	65.12	72.16	91.37	64.64	94.32	75.69	31.96	93.28	69.52	85.33	65.92	63.21	75.24	73.00
Unfreeze 1	97.75	82.13	91.70	60.25	55.13	53.37	64.63	72.45	91.38	65.41	94.36	79.52	56.15	93.62	71.50	85.13	68.28	58.96	75.28	74.58
Unfreeze 2	97.76	82.20	91.77	60.22	57.63	54.64	65.50	73.82	91.52	64.60	94.37	80.69	59.61	94.07	75.97	83.71	65.35	61.37	76.09	75.31
Unfreeze 3	97.96	83.33	92.20	61.45	61.32	56.89	66.19	75.54	91.69	65.02	94.42	81.34	61.27	94.30	76.32	84.80	67.51	58.73	76.09	76.13

returns catastrophic FPR95 rates across almost all EoMT states.

Finally, although the zero-shot COCO model fails at anomaly isolation due to poor urban alignment, our fine-tuning strategy restores this capacity. Combined with Max-Entropy, it yields the strongest overall result on SMIYC RA-21: 91.74% AuPRC and 2.60% FPR95.

3.4. Temperature Scaling

Table 4 isolates the effects of Temperature Scaling (T) on the fine-tuned EoMT model with MSP. Exploring a range of temperatures reveals a strict dataset-dependent trade-off: lower values (e.g., $T = 0.5$) boost AuPRC on discrete anomalies but severely inflate false positives on SMIYC RA-21 (reaching 34.04% FPR95), whereas higher values yield diminishing returns. Evaluating these trends, we empirically fix $T = 0.7$ as a balanced trade-off, preserving detection gains without sacrificing baseline stability across broader scenes.

4. Discussion

4.1. Zero-Shot Limitations and Fine-Tuning

The significant performance drop on urban infrastructure during the zero-shot evaluation shows a key limitation of the COCO-pretrained model. Since the network was originally trained on generic images, it learned to recognize basic object shapes but lacks the specific structural details needed to understand a complex driving scene. This explains why it completely misses specialized urban elements. Our fine-tuning results reveal how the progressive unfreezing strategy actually bridges this domain gap. If we only update the final prediction head, the model simply reorganizes the features it already knows, which is not enough to recognize new, unmapped categories. However, by unfreezing the deeper layers of the encoder, we allow the model to actively learn new patterns and adapt its internal features to the urban environment.

4.2. Comparison of Pixel-Level and Mask-Level Anomalies

The pixel-based ERFNet yields independent predictions due to its localized receptive field, causing local overconfidence on OoD textures and producing false positives dominated by object boundaries (visible as noisy edges in Fig. 2). Conversely, mask-based architectures leverage object-level queries to group regions into structural entities. While the domain-aligned EoMT Cityscapes effectively isolates the anomaly, the zero-shot EoMT COCO fails because its generic training taxonomy includes animals, leading it to treat the OoD object as in-distribution. Finally, our fine-tuned EoMT COCO successfully overrides this bias, returning a sharply defined mask that uniformly highlights the anomaly without edge-level artifacts (Fig. 2, rightmost panel).

4.3. Evaluation of Post-Hoc Metrics

The interaction between mask-based outputs and post-hoc heuristics governs the metric volatility in Table 3. MaxEntropy's exceptional performance on the fine-tuned EoMT is driven by the asymptotic properties of Softmax and Sigmoid activations. For OoD regions, known-class masks typically

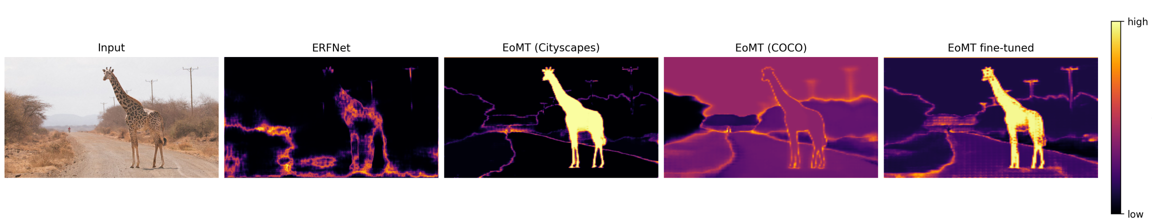

Figure 2. Anomaly maps on RoadAnomaly21 with MSP. Left to right: input, ERFNet, EoMT Cityscapes, zero-shot COCO, and fine-tuned COCO.

Table 3. Anomaly segmentation results. AuPRC is higher; FPR95 is lower.

Model	Method	mIoU_{17}	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
ERFNet	MSP	73.93	29.10	62.55	2.71	65.22	1.75	50.60	7.47	41.84	12.42	82.58
ERFNet	MaxLogit	73.93	38.32	59.34	4.63	48.44	3.30	45.49	9.50	40.30	15.58	73.25
ERFNet	MaxEntropy	73.93	30.97	62.66	3.04	65.91	2.58	50.16	8.84	41.55	12.67	82.75
EoMT Cityscapes	MSP	83.60	75.97	34.53	89.76	0.44	27.15	12.07	55.76	35.78	77.03	19.50
EoMT Cityscapes	MaxLogit	83.60	75.09	20.96	47.56	100.00	28.03	12.59	55.72	73.67	73.36	41.94
EoMT Cityscapes	MaxEntropy	83.60	68.21	27.34	85.39	0.73	6.44	29.99	12.94	53.82	52.09	45.22
EoMT Cityscapes	RbA	83.60	72.07	79.12	85.61	99.91	27.86	7.69	57.18	80.95	76.60	19.72
EoMT COCO	MSP	62.92	35.59	87.79	1.00	99.99	1.70	96.62	7.08	96.61	21.49	94.28
EoMT COCO	MaxLogit	62.92	28.91	75.84	28.50	100.00	0.48	91.33	8.16	74.53	18.64	85.78
EoMT COCO	MaxEntropy	62.92	21.81	89.57	15.57	100.00	1.48	38.44	6.59	94.85	16.87	93.77
EoMT COCO	RbA	62.92	23.64	64.67	38.85	99.99	0.20	96.68	1.51	99.96	17.77	79.44
EoMT COCO-FT	MSP	79.20	80.13	15.38	62.99	1.66	31.89	20.25	77.63	13.45	70.86	27.73
EoMT COCO-FT	MaxLogit	79.20	34.83	41.17	63.74	21.24	6.14	48.92	45.43	45.42	59.74	27.59
EoMT COCO-FT	MaxEntropy	79.20	91.74	2.60	85.50	6.77	12.18	34.87	35.98	61.73	68.53	35.69
EoMT COCO-FT	RbA	79.20	43.92	92.83	50.25	99.94	30.96	91.50	80.88	76.43	68.18	71.09

Table 4. Temperature scaling on fine-tuned EoMT with MSP.

Method	mIoU_{17}	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP	79.20	80.13	15.38	62.99	1.66	31.89	20.25	77.63	13.45	70.86	27.73
MSP ($T = 0.5$)	79.20	76.99	34.04	76.72	1.45	35.98	18.72	77.16	12.24	77.28	36.08
MSP ($T = 0.75$)	79.20	78.88	21.22	70.81	1.45	34.12	18.56	77.40	13.01	74.14	29.83
MSP ($T = 1.1$)	79.20	80.41	13.93	60.08	1.99	30.99	20.50	77.79	13.47	70.03	26.73
MSP (best $T = 0.7$)	79.20	78.56	22.93	72.08	1.45	34.56	18.63	77.27	12.93	75.04	30.29

produce low-confidence responses distributed across multiple classes. After aggregation and renormalization, the resulting class distribution tends to become flatter, leading to higher entropy values. Empirically, this allows entropy-based scoring to better capture uncertain or ambiguous regions compared to raw logit-based metrics. Conversely, MaxLogit and RbA fail on EoMT due to uncalibrated raw mask logits. Unlike pixel-based networks where raw logits provide reliable energy boundaries, mask queries output unbounded spatial-semantic scores. Aggregating these values without normalization constraints produces volatile anomaly scores that destroy the False Positive Rate, rendering them unsuitable for practical deployment.

4.4. Limitations and Practical Challenges

While effective, our methodology exposes practical limitations. First of all, a vulnerability of mask-based OoD detection consists in its structural ambiguity: new objects structurally similar to known classes (e.g., animals mimicking pedestrians) can erroneously trigger high confidence for those categories, masking the anomaly. Regarding the calibration, the temperature T is strongly sensitive to the dataset. For this reason, relying on a fixed value is not safe for autonomous driving, where obstacle detection is a critical component. Finally, the transition from a lightweight CNN (*ERFNet*) to a Transformer-based mask architecture (*EoMT*) introduces severe latency, which represents a critical bottleneck for real-time (high FPS) safety systems.

5. Conclusions

In this work, we evaluated the EoMT architecture for anomaly detection in road scenes. To address the domain gap between generic COCO and the specialized Cityscapes environment, we implemented a custom mapping function and fine-tuned the model. Regarding out-of-distribution detection, we observed that while region-level metrics like RbA seem promising for mask classification, they actually lead to overconfident and poorly calibrated predictions. Instead, applying MaxEntropy directly on mask queries proved much more effective, outperforming the baselines on the SMIYC and Fishyscapes datasets. Then, we applied temperature scaling to improve the calibration of the confidence scores. The main limitation of our approach remains its strong reliance on in-distribution pre-training. Future works could address this vulnerability by using anomaly synthesis during training, forcing the model to explicitly learn out-of-distribution features for safer real-world deployment.

6. Acknowledgments

We declare that Generative AI tools were not used to write the text of this report.

Gemini 3.1 Pro (Google) was employed solely to assist with code drafting, debugging, and development planning. The final implementation, experimental setup, and data analysis were fully managed and verified by the authors.

References

[1] Hermann Blum, Paul-Edouard Sarlin, Juan Nieto, Roland Siegwart, and Cesar Cadena. The fishyscapes benchmark: Measuring blind spots in semantic segmentation. *International Journal of Computer Vision*, 129(11):3119–3135, 2021. 1

[2] Robin Chan, Krzysztof Lis, Svenja Uhlemeyer, Hermann Blum, Sina Honari, Roland Siegwart, Mathieu Salzmann, Pascal Fua, and Matthias Rottmann. Segmentmeifyoucan: A benchmark for anomaly segmentation. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2021. 1

[3] Bowen Cheng, Alexander Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 17864–17875, 2021. 1

[4] Bowen Cheng, Ishan Misra, Alexander G. Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation, 2022. 1

[5] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 1

[6] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks, 2018. 2

[7] Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joe Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling out-of-distribution detection for real-world settings, 2022. 2

[8] Tommie Kerssies, Niccolò Cavagnero, Alexander Hermans, Narges Norouzi, Giuseppe Averta, Bastian Leibe, Gijs Dubbelman, and Daan de Geus. Your ViT is secretly an image segmentation model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 1, 2

[9] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer Vision–ECCV 2014*, pages 740–755. Springer, 2014. 1

[10] Nazir Nayal, Mısra Yavuz, João F. Henriques, and Fatma Güney. Rba: Segmenting unknown regions rejected by all, 2023. 1, 2, 3

[11] Maxime Oquab, Timothée Darcet, Theo Moutakanni, Huy V Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. 1

[12] Eduardo Romera, José M Alvarez, Luis M Bergasa, and Roberto Arroyo. Erfnet: Efficient residual factorized convnet for real-time semantic segmentation. *IEEE Transactions on Intelligent Transportation Systems*, 19(1):263–272, 2017. 1